PRÁCTICA PREDICTIVA MINERIA DE BASES DE DATOS

CRISTIAN CAMILO BRAN ARRIAGA
SANTIAGO CARDONA VILLADA
RICHARD JAVIER MORENO MOSQUERA

ESPECIALIZACIÓN DE BIG DATA E INTELIGENCIA DE NEGOCIOS

UNIVERSIDAD CATÓLICA LUISAMIGÓ

MEDELLIN

2025

Objetivo de la minería

Predecir el tipo de fármaco (drug) que se debe administrar a un paciente afectado de rinitis alérgica según distintos parámetros/variables. Se han recogido los datos del medicamento idóneo para muchos pacientes en cuatro hospitales. Se pretende, para nuevos pacientes, determinar el mejor medicamento a probar.

Descripción de los datos

Las variables disponibles en los historiales clínicos de cada paciente son:

• **Age**: Edad

• **Sex**: Sexo

• **BP** (Blood Pressure): Tension Sanguínea.

• **Cholesterol**: nivel de colesterol.

• **Na**: Nivel de sodio en la sangre.

• **K**: Nivel de potasio en la sangre.

• **Drug**: Hay cinco fármacos posibles {DrugA, DrugB, DrugC, DrugX, DrugY}

Modelamiento

Se requiere:

1. Preparar los datos para el análisis

2. Aplicar todos los métodos vistos en clase y analizar cada resultado obtenido por WEKA. Hacer un informe indicando los parámetros óptimos utilizados en cada método y los resultados obtenidos.

3. Establecer un criterio para seleccionar el método que mejor se ajuste a las necesidades del problema. Justificar la elección.

4. Utilice la pestaña paciente_nuevos para realizar la validación del modelo.

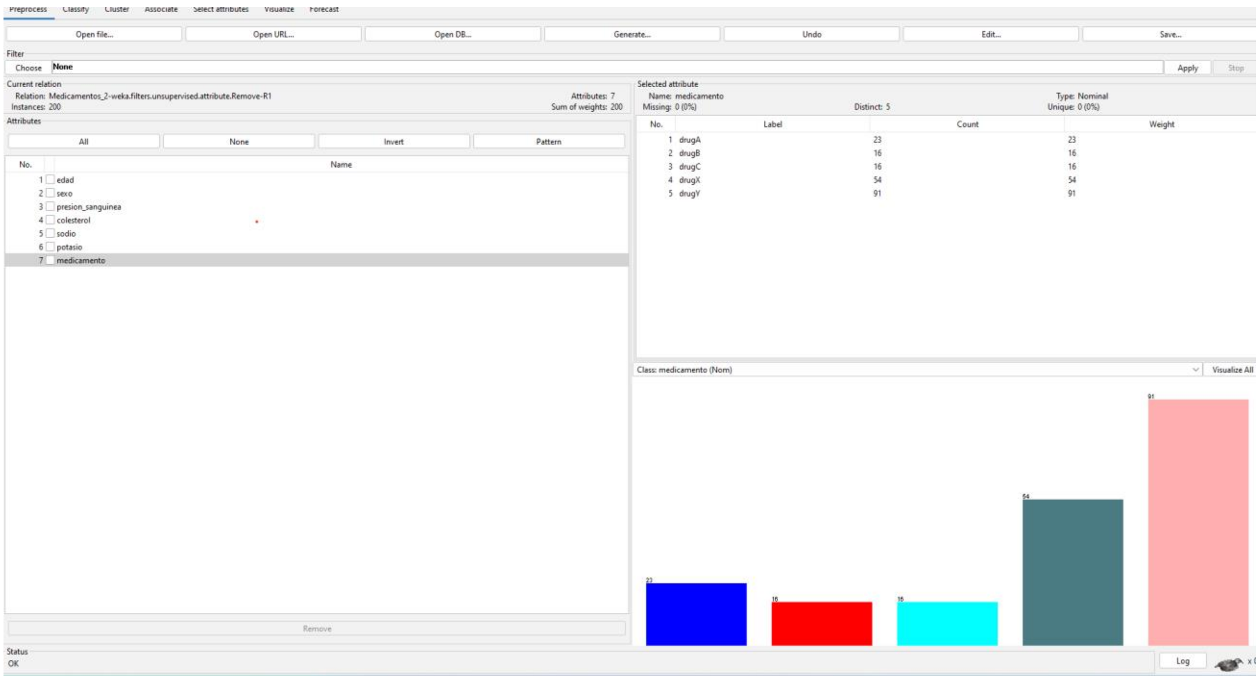
Excel_Drug

Bloc de notas Medicamentos_2

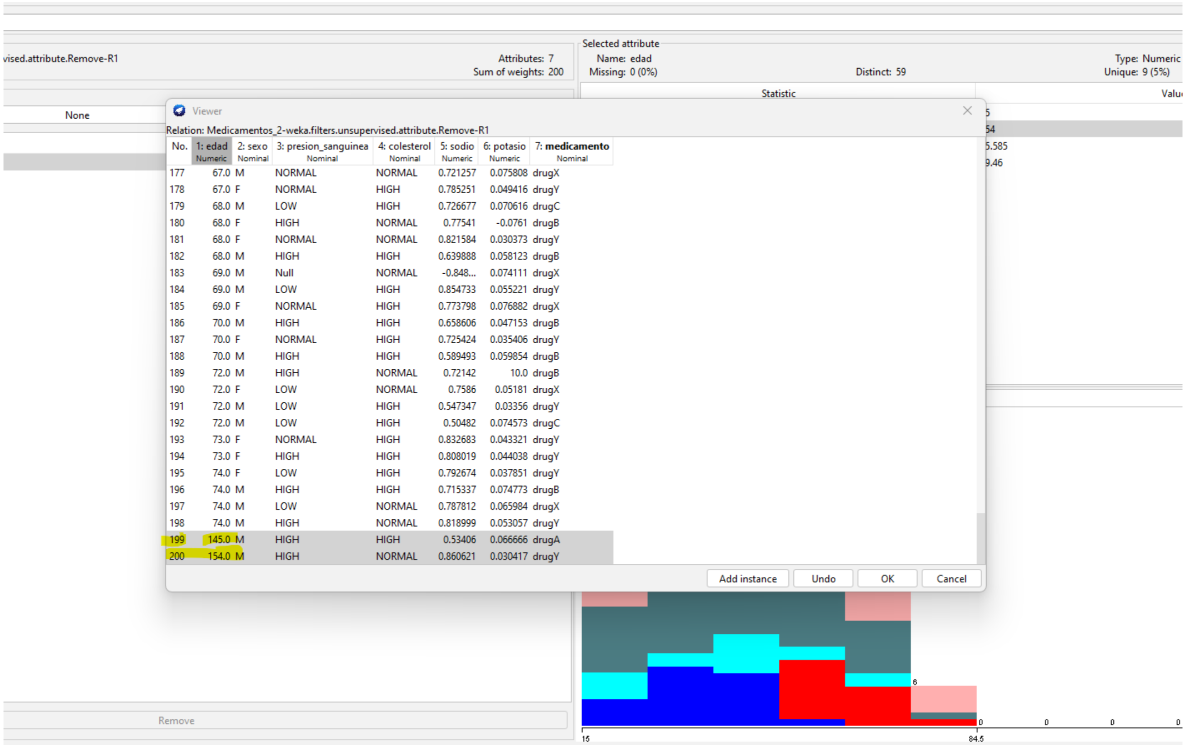
Bloc de notas Pacientes_pruebas

Resultados:

1. Se inicia eliminando la variable “**id**” por ser un dato irrelevante.

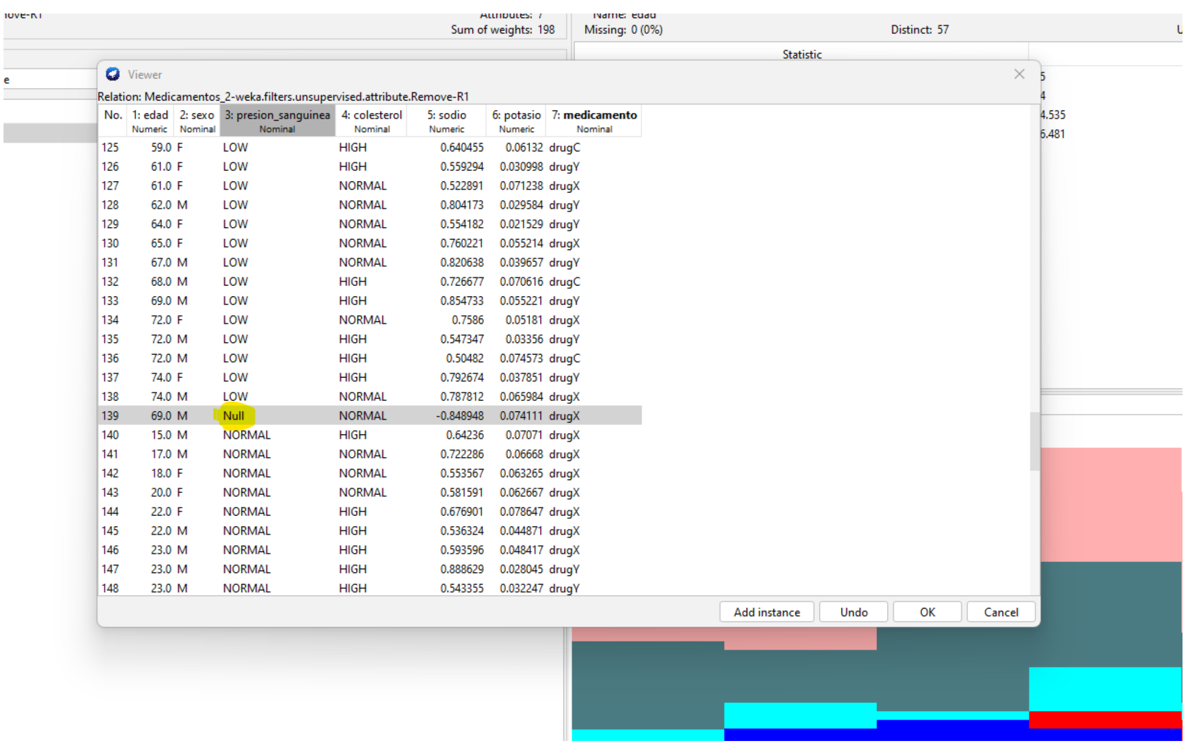


Eliminación de datos atípicos en “**edad”** y “**presión sanguínea”.**



Relation: Medicamentos_2-weka.filters.unsupervised.attribute.Remove-R1

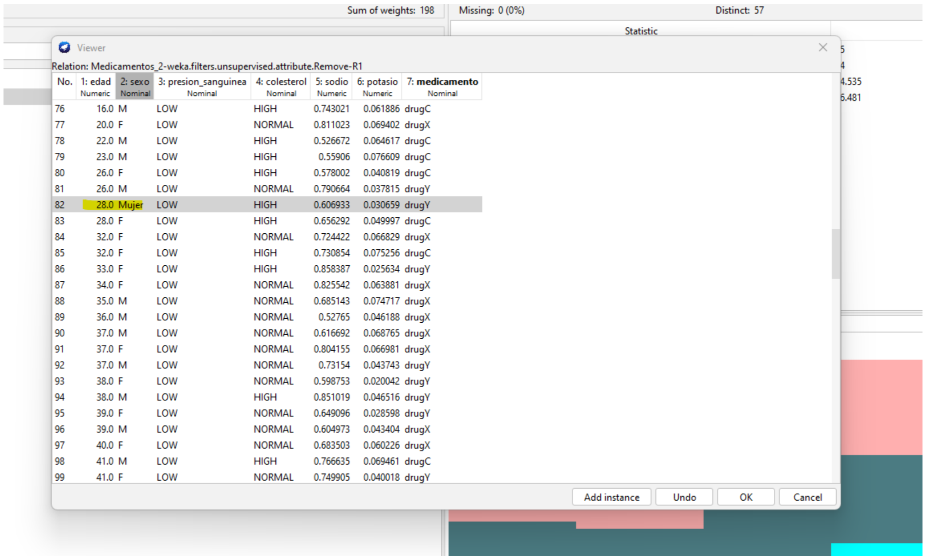
No.	1: edad (Numeric)	2: sexo (Nominal)	3: presion_sanguinea (Nominal)	4: colesterol (Nominal)	5: sodio (Numeric)	6: potasio (Numeric)	7: medicamento (Nominal)
177	67.0	M	NORMAL	NORMAL	0.721257	0.075808	drugX
178	67.0	F	NORMAL	HIGH	0.785251	0.049416	drugY
179	68.0	M	LOW	HIGH	0.726677	0.070616	drugC
180	68.0	F	HIGH	NORMAL	0.77541	-0.0761	drugB
181	68.0	F	NORMAL	NORMAL	0.821584	0.030373	drugY
182	68.0	M	HIGH	HIGH	0.639888	0.058123	drugB
183	69.0	M	Null	NORMAL	-0.848...	0.074111	drugX
184	69.0	M	LOW	HIGH	0.854733	0.055221	drugY
185	69.0	F	NORMAL	HIGH	0.773798	0.076882	drugX
186	70.0	M	HIGH	HIGH	0.658606	0.047153	drugB
187	70.0	F	NORMAL	HIGH	0.725424	0.035406	drugY
188	70.0	M	HIGH	HIGH	0.589493	0.059854	drugB
189	72.0	M	HIGH	NORMAL	0.72142	10.0	drugB
190	72.0	F	LOW	NORMAL	0.7586	0.05181	drugX
191	72.0	M	LOW	HIGH	0.547347	0.03356	drugY
192	72.0	M	LOW	HIGH	0.50482	0.074573	drugC
193	73.0	F	NORMAL	HIGH	0.832683	0.043321	drugY
194	73.0	F	HIGH	HIGH	0.808019	0.044038	drugY
195	74.0	F	LOW	HIGH	0.792674	0.037851	drugY
196	74.0	M	HIGH	HIGH	0.715337	0.074773	drugB
197	74.0	M	LOW	NORMAL	0.787812	0.065984	drugX
198	74.0	M	HIGH	NORMAL	0.818999	0.053057	drugY
199	145.0	M	HIGH	HIGH	0.53406	0.066666	drugA
200	154.0	M	HIGH	NORMAL	0.860621	0.030417	drugY



Relation: Medicamentos_2-weka.filters.unsupervised.attribute.Remove-R1

No.	1: edad (Numeric)	2: sexo (Nominal)	3: presion_sanguinea (Nominal)	4: colesterol (Nominal)	5: sodio (Numeric)	6: potasio (Numeric)	7: medicamento (Nominal)
125	59.0	F	LOW	HIGH	0.640455	0.06132	drugC
126	61.0	F	LOW	HIGH	0.559294	0.030998	drugY
127	61.0	F	LOW	NORMAL	0.522891	0.071238	drugX
128	62.0	M	LOW	NORMAL	0.804173	0.029584	drugY
129	64.0	F	LOW	NORMAL	0.554182	0.021529	drugY
130	65.0	F	LOW	NORMAL	0.760221	0.055214	drugX
131	67.0	M	LOW	NORMAL	0.820638	0.039657	drugY
132	68.0	M	LOW	HIGH	0.726677	0.070616	drugC
133	69.0	M	LOW	HIGH	0.854733	0.055221	drugY
134	72.0	F	LOW	NORMAL	0.7586	0.05181	drugX
135	72.0	M	LOW	HIGH	0.547347	0.03356	drugY
136	72.0	M	LOW	HIGH	0.50482	0.074573	drugC
137	74.0	F	LOW	HIGH	0.792674	0.037851	drugY
138	74.0	M	LOW	NORMAL	0.787812	0.065984	drugX
139	69.0	M	Null	NORMAL	-0.848948	0.074111	drugX
140	15.0	M	NORMAL	HIGH	0.64236	0.07071	drugX
141	17.0	M	NORMAL	NORMAL	0.722286	0.06668	drugX
142	18.0	F	NORMAL	NORMAL	0.553567	0.063265	drugX
143	20.0	F	NORMAL	NORMAL	0.581591	0.062667	drugX
144	22.0	F	NORMAL	HIGH	0.676901	0.078647	drugX
145	22.0	M	NORMAL	HIGH	0.536324	0.044871	drugX
146	23.0	M	NORMAL	HIGH	0.593596	0.048417	drugX
147	23.0	M	NORMAL	HIGH	0.888629	0.028045	drugY
148	23.0	M	NORMAL	HIGH	0.543355	0.032247	drugY

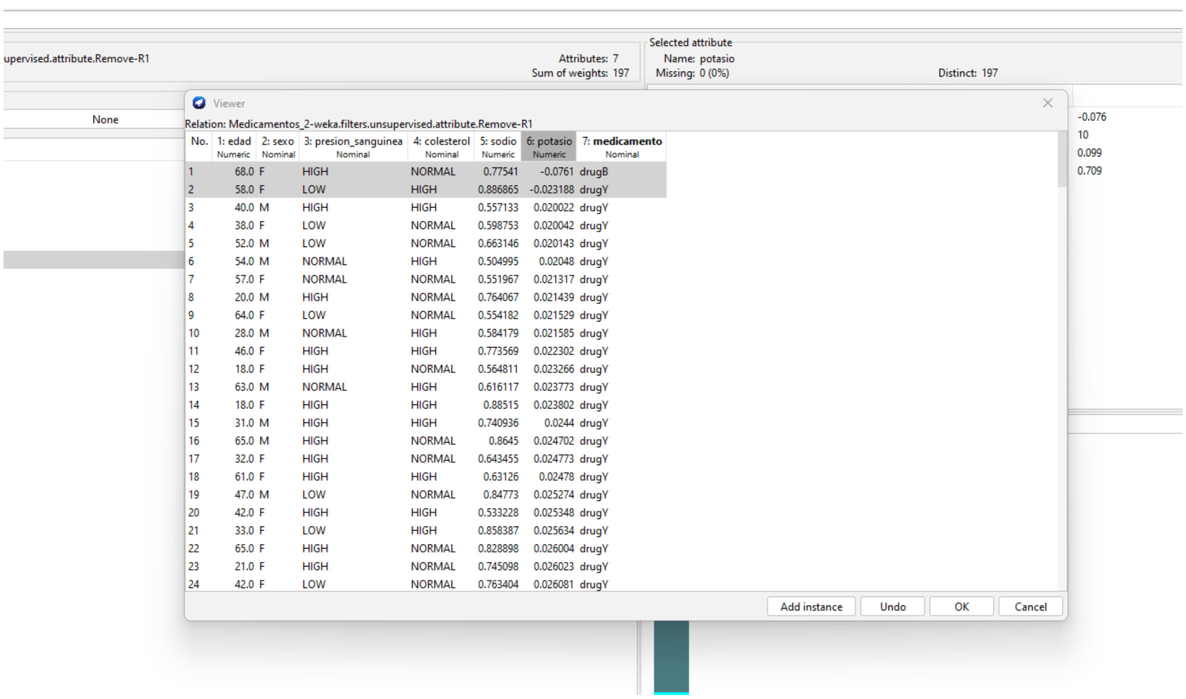
Se realiza el cambio de dato “**Mujer”** a “**M”.**



Relation: Medicamentos_2-weka.filters.unsupervised.attribute.Remove-R1

No.	1: edad Numeric	2: sexo Nominal	3: presion_sanguinea Nominal	4: colesterol Nominal	5: sodio Numeric	6: potasio Numeric	7: medicamento Nominal
76	16.0	M	LOW	HIGH	0.743021	0.061886	drugC
77	20.0	F	LOW	NORMAL	0.811023	0.069402	drugX
78	22.0	M	LOW	HIGH	0.526672	0.064617	drugC
79	23.0	M	LOW	HIGH	0.55906	0.076609	drugC
80	26.0	F	LOW	HIGH	0.578002	0.040819	drugC
81	26.0	M	LOW	NORMAL	0.790664	0.037815	drugY
82	28.0	Mujer	LOW	HIGH	0.606933	0.030659	drugY
83	28.0	F	LOW	HIGH	0.656292	0.049997	drugC
84	32.0	F	LOW	NORMAL	0.724422	0.066829	drugX
85	32.0	F	LOW	HIGH	0.730854	0.075256	drugC
86	33.0	F	LOW	HIGH	0.858387	0.025634	drugY
87	34.0	F	LOW	NORMAL	0.825542	0.063881	drugX
88	35.0	M	LOW	NORMAL	0.685143	0.074717	drugX
89	36.0	M	LOW	NORMAL	0.52765	0.046188	drugX
90	37.0	M	LOW	NORMAL	0.616692	0.068765	drugX
91	37.0	F	LOW	NORMAL	0.804155	0.066981	drugX
92	37.0	M	LOW	NORMAL	0.73154	0.043743	drugY
93	38.0	F	LOW	NORMAL	0.598753	0.020042	drugY
94	38.0	M	LOW	HIGH	0.851019	0.046516	drugY
95	39.0	F	LOW	NORMAL	0.649096	0.028598	drugY
96	39.0	M	LOW	NORMAL	0.604973	0.043404	drugX
97	40.0	F	LOW	NORMAL	0.683503	0.060226	drugX
98	41.0	M	LOW	HIGH	0.766635	0.069461	drugC
99	41.0	F	LOW	NORMAL	0.749905	0.040018	drugY

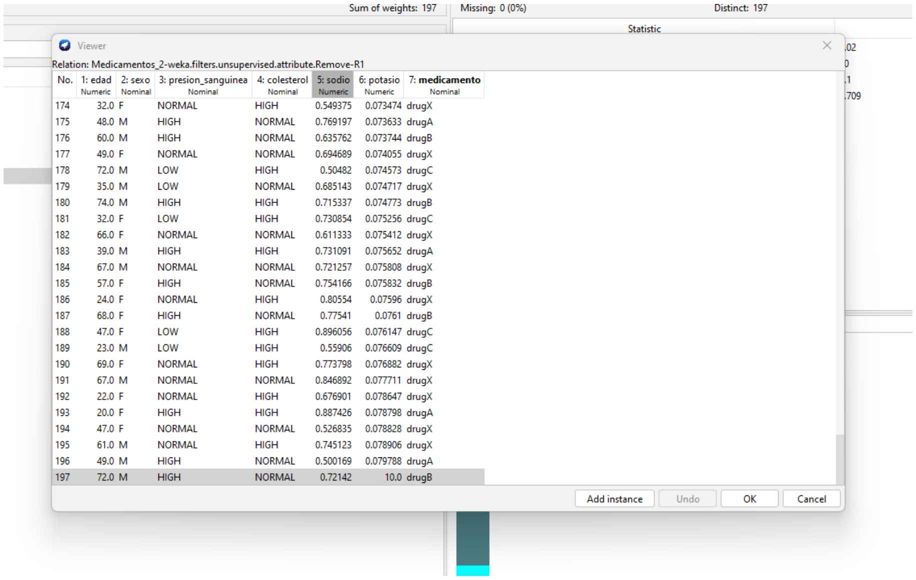
Se cambian los datos de “**Potasio**” que se encontraban en estado negativo.



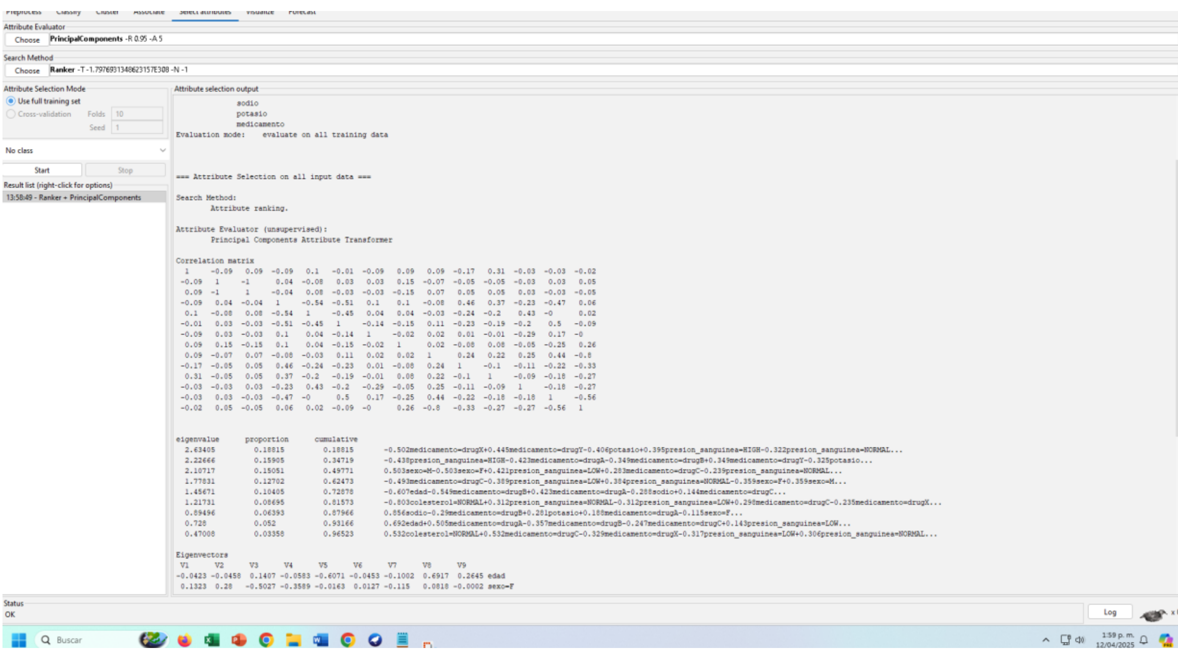
Relation: Medicamentos_2-weka.filters.unsupervised.attribute.Remove-R1

No.	1: edad Numeric	2: sexo Nominal	3: presion_sanguinea Nominal	4: colesterol Nominal	5: sodio Numeric	6: potasio Numeric	7: medicamento Nominal
1	68.0	F	HIGH	NORMAL	0.77541	-0.0761	drugB
2	58.0	F	LOW	HIGH	0.886865	-0.023188	drugY
3	40.0	M	HIGH	HIGH	0.557133	0.020022	drugY
4	38.0	F	LOW	NORMAL	0.598753	0.020042	drugY
5	52.0	M	LOW	NORMAL	0.663146	0.020143	drugY
6	54.0	M	NORMAL	HIGH	0.504995	0.02048	drugY
7	57.0	F	NORMAL	NORMAL	0.551967	0.021317	drugY
8	20.0	M	HIGH	NORMAL	0.764067	0.021439	drugY
9	64.0	F	LOW	NORMAL	0.554182	0.021529	drugY
10	28.0	M	NORMAL	HIGH	0.584179	0.021585	drugY
11	46.0	F	HIGH	HIGH	0.773569	0.022302	drugY
12	18.0	F	HIGH	NORMAL	0.564811	0.023266	drugY
13	63.0	M	NORMAL	HIGH	0.616117	0.023773	drugY
14	18.0	F	HIGH	HIGH	0.88515	0.023802	drugY
15	31.0	M	HIGH	HIGH	0.740936	0.0244	drugY
16	65.0	M	HIGH	NORMAL	0.8645	0.024702	drugY
17	32.0	F	HIGH	NORMAL	0.643455	0.024773	drugY
18	61.0	F	HIGH	HIGH	0.63126	0.02478	drugY
19	47.0	M	LOW	NORMAL	0.84773	0.025274	drugY
20	42.0	F	HIGH	HIGH	0.533228	0.025348	drugY
21	33.0	F	LOW	HIGH	0.858387	0.025634	drugY
22	65.0	F	HIGH	NORMAL	0.828898	0.026004	drugY
23	21.0	F	HIGH	NORMAL	0.745098	0.026023	drugY
24	42.0	F	LOW	NORMAL	0.763404	0.026081	drugY

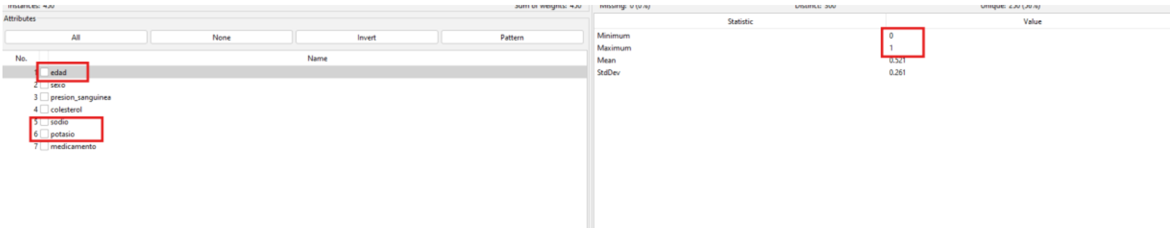
Se realiza eliminación de un dato atípico en Potasio con valor 10.



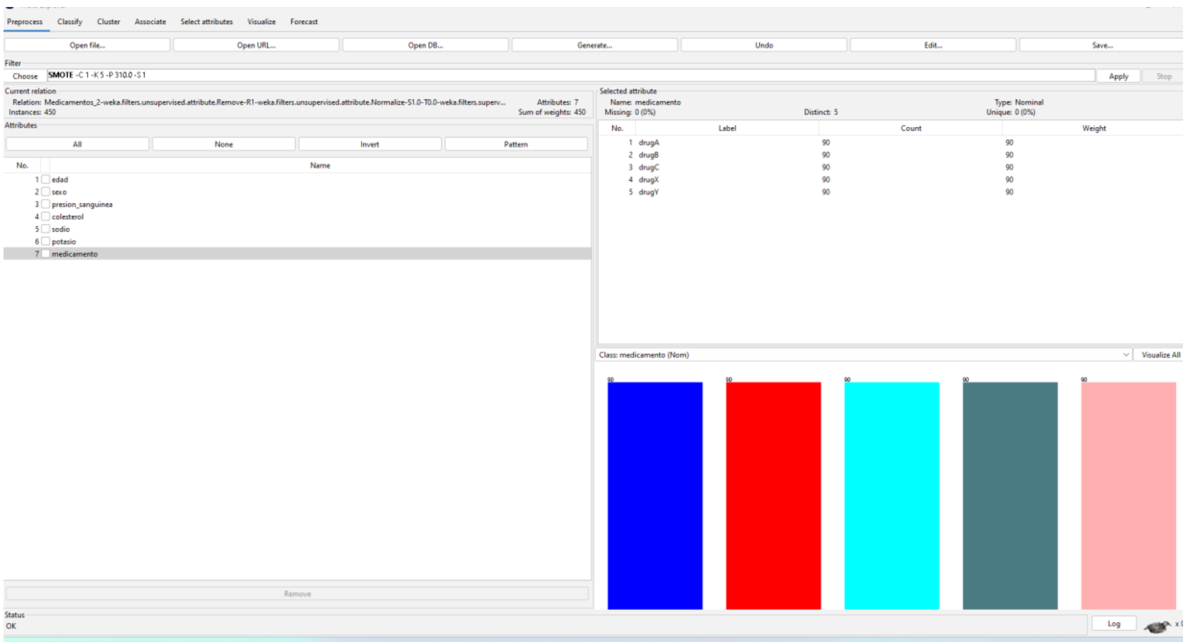
En el análisis de correlación no hay variables con correlaciones por encima de 0,7.



Luego se realiza normalización de datos en rangos de 0-1.



Balanceo hacia arriba.



2. Método de árbol de decisión:

Se encuentra luego de realizar diferentes parametrizaciones, un error bajo del 0.97, en la matriz de confusión solo se tienen dos falsos positivos, lo que da un buen balance.

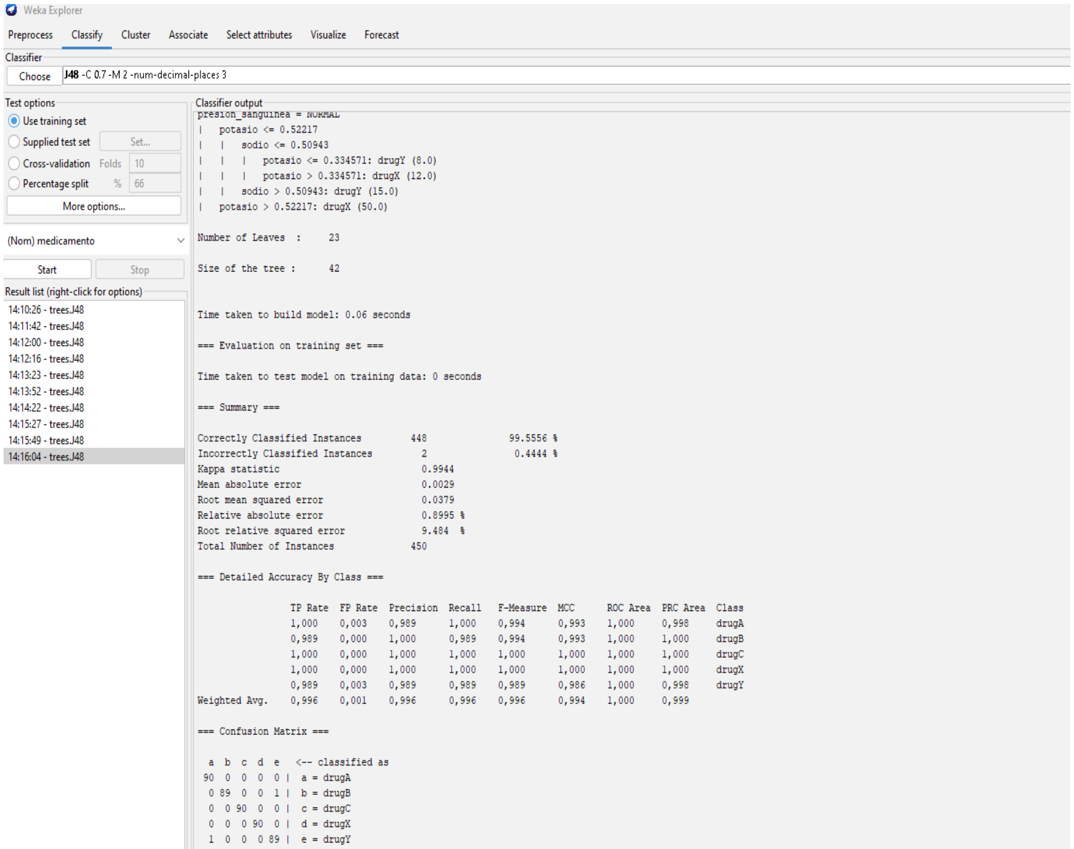
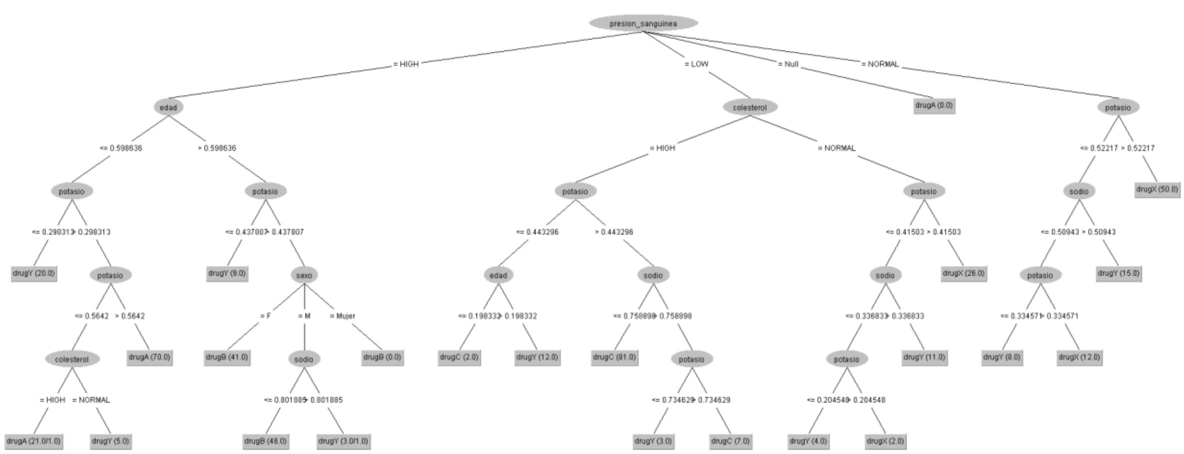
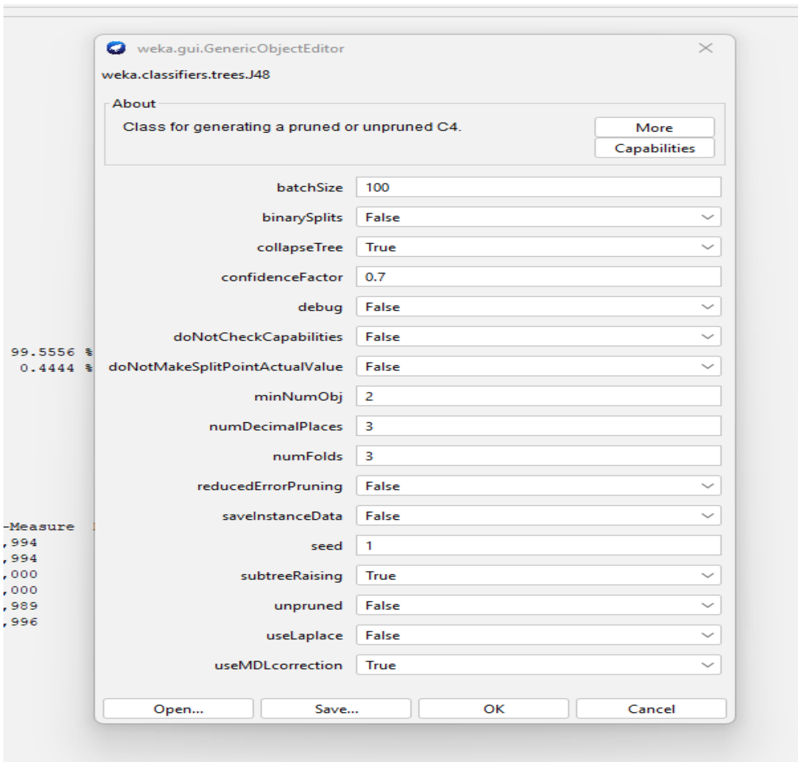


Gráfico de árbol de decisiones

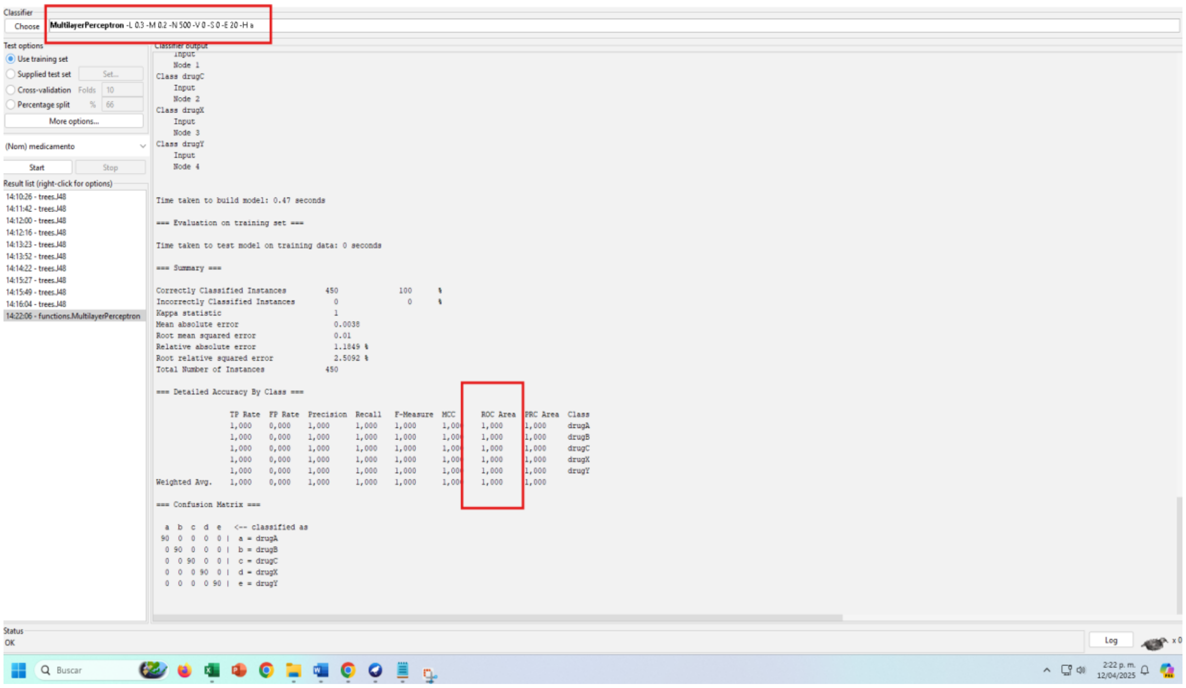


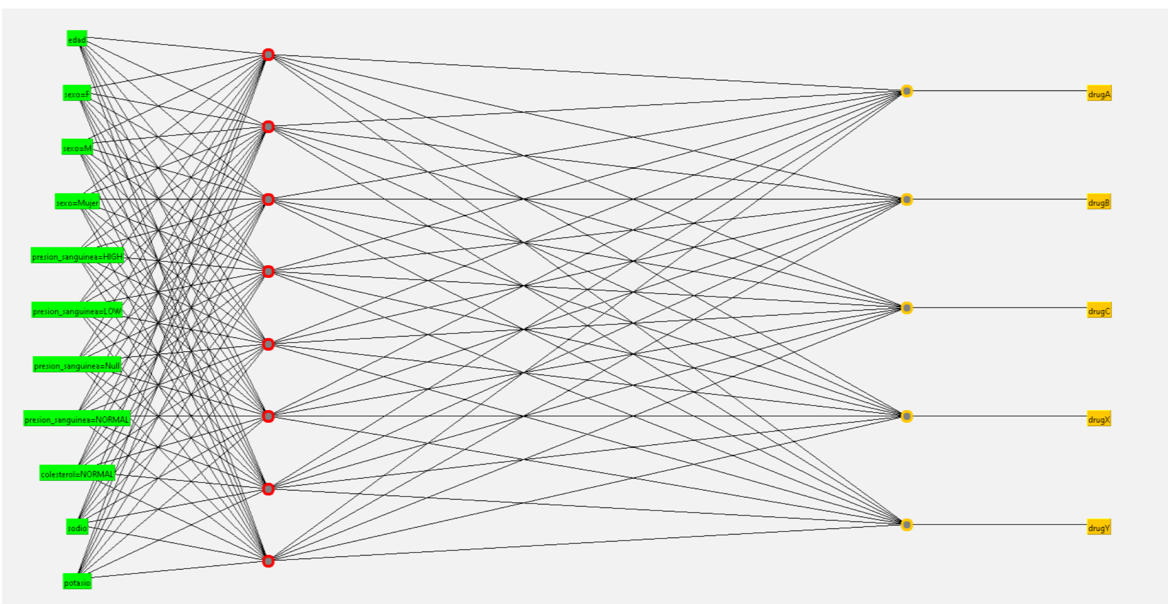
Configuración de clasificación



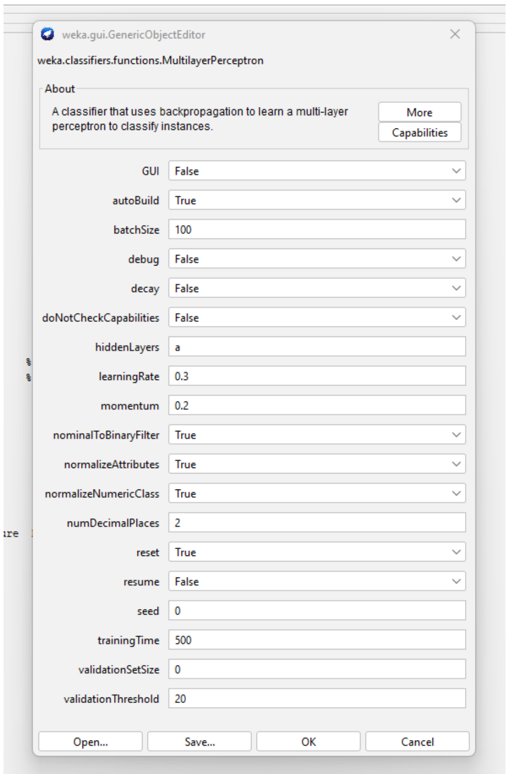
Método de redes neuronales:

Se realiza el análisis con la red neuronal y se encuentra un resultado positivo.


Classifier
Choose
MultilayerPerceptron -L 0.3 -M 0.2 -N 500 -V 0 -S 0 -E 20 -H a
Test options
Use training set
Supplied test set
Set...
Cross-validation Folds 10
Percentage split % 66
More options...
(Nom) medicamento
Start
Stop
Result list (right-click for options)
14:10:26 - trees.J48
14:11:42 - trees.J48
14:12:00 - trees.J48
14:12:16 - trees.J48
14:13:23 - trees.J48
14:13:52 - trees.J48
14:14:22 - trees.J48
14:15:27 - trees.J48
14:15:49 - trees.J48
14:16:04 - trees.J48
14:22:06 - functions.MultilayerPerceptron
Classifier output
Input
Node 1
Class drugC
Input
Node 2
Class drugX
Input
Node 3
Class drugY
Input
Node 4
Time taken to build model: 0.47 seconds
=== Evaluation on training set ===
Time taken to test model on training data: 0 seconds
=== Summary ===
Correctly Classified Instances 450 100 %
Incorrectly Classified Instances 0 0 %
Kappa statistic 1
Mean absolute error 0.0038
Root mean squared error 0.01
Relative absolute error 1.1849 %
Root relative squared error 2.5092 %
Total Number of Instances 450
=== Detailed Accuracy By Class ===
TP Rate FP Rate Precision Recall F-Measure MCC ROC Area PRC Area Class
1,000 0,000 1,000 1,000 1,000 1,000 1,000 1,000 drugA
1,000 0,000 1,000 1,000 1,000 1,000 1,000 1,000 drugB
1,000 0,000 1,000 1,000 1,000 1,000 1,000 1,000 drugC
1,000 0,000 1,000 1,000 1,000 1,000 1,000 1,000 drugX
1,000 0,000 1,000 1,000 1,000 1,000 1,000 1,000 drugY
Weighted Avg. 1,000 0,000 1,000 1,000 1,000 1,000 1,000 1,000
=== Confusion Matrix ===
a b c d e <-- classified as
90 0 0 0 0 | a = drugA
0 90 0 0 0 | b = drugB
0 0 90 0 0 | c = drugC
0 0 0 90 0 | d = drugX
0 0 0 0 90 | e = drugY
Status
OK
Log
Buscar
2:22 p. m.
12/04/2025

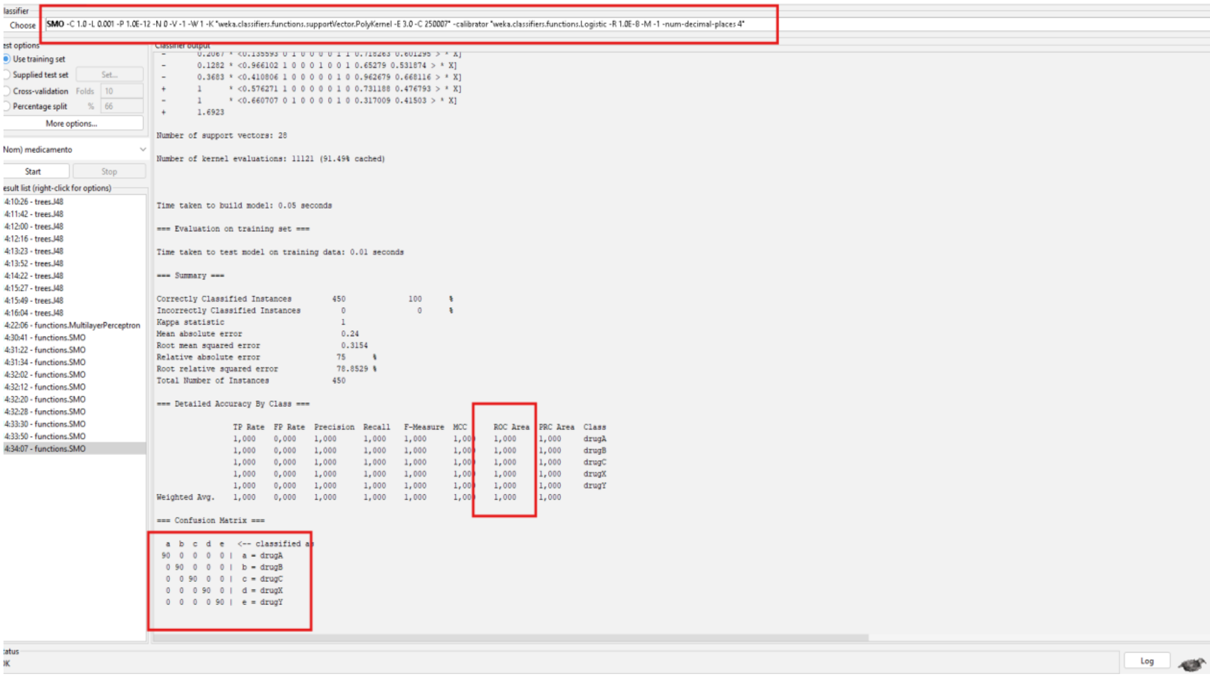

edad
sexo=F
sexo=M
sexo=Mujer
presion_sanguinea=HIGH
presion_sanguinea=LOW
presion_sanguinea=Null
presion_sanguinea=NORMAL
colesterol=NORMAL
sodio
potasio
drugA
drugB
drugC
drugX
drugY

Configuración de clasificación.

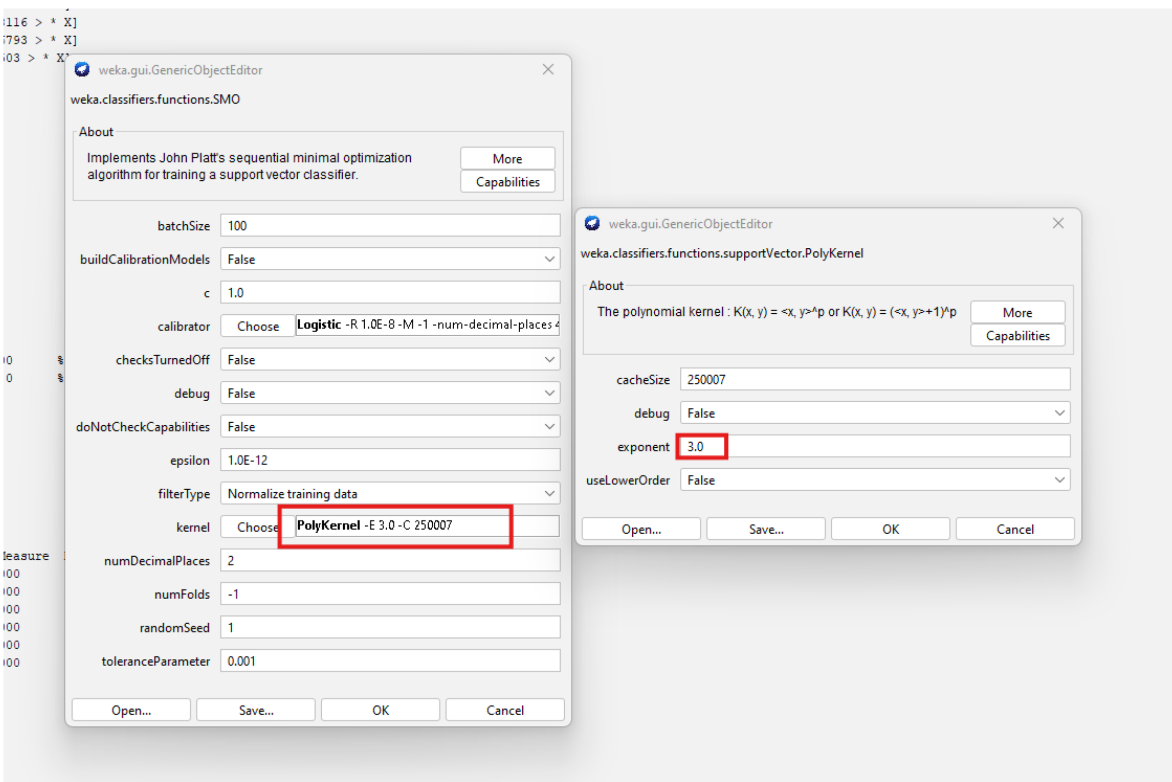


Máquinas de soporte vectorial:

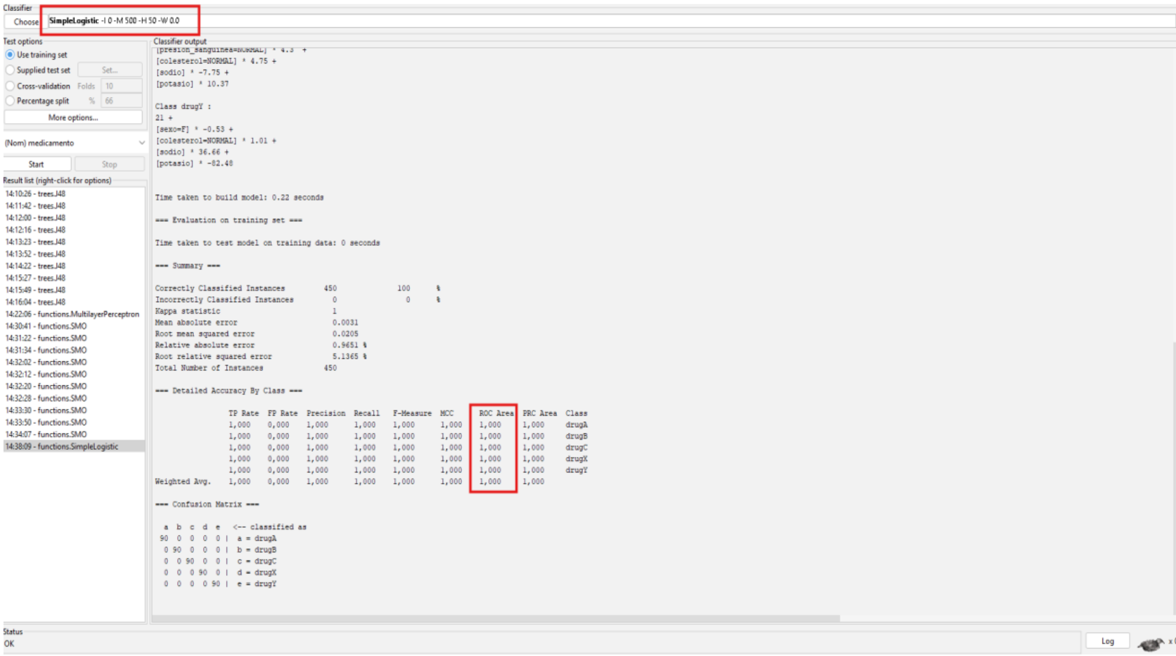
Se encuentra un resultado optimo



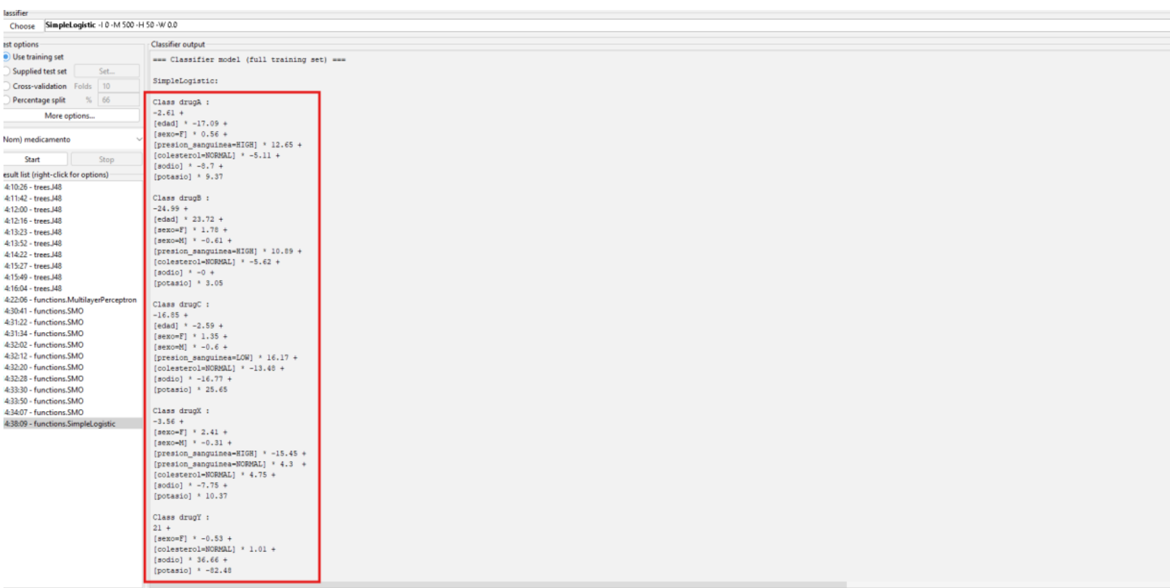
Configuración de clasificación



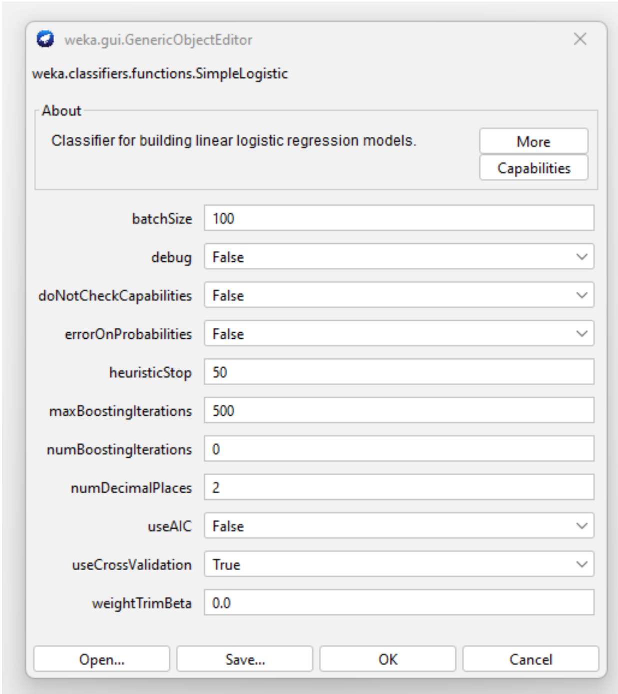
Regresión



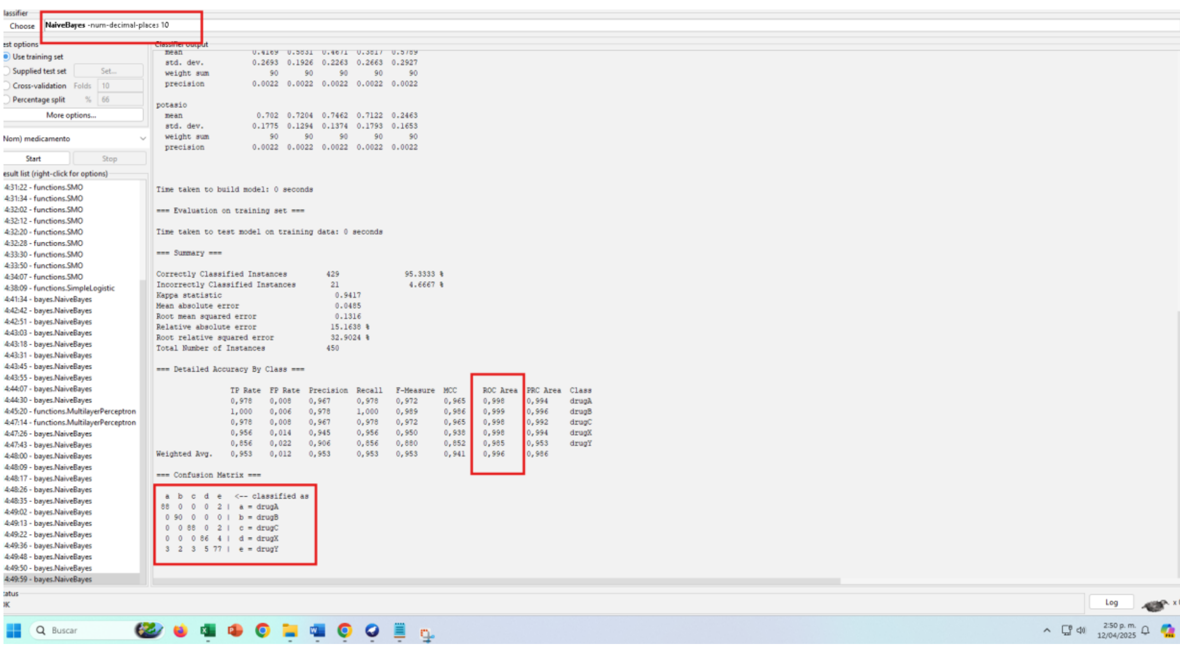
Clases resultado



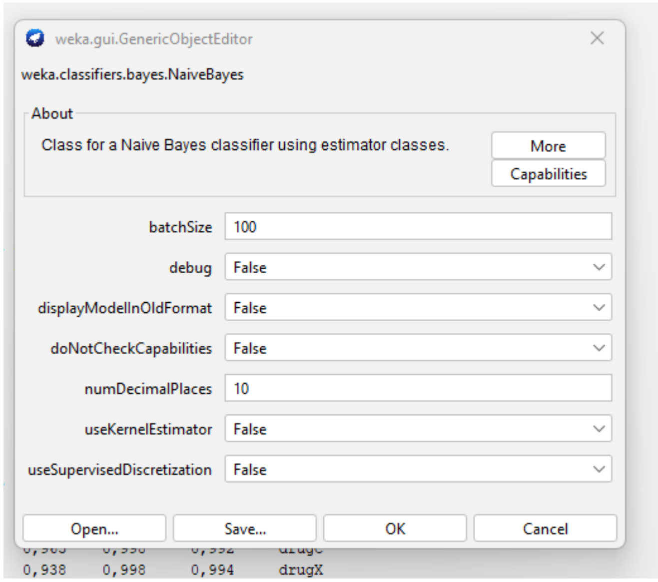
Configuración de caracterización



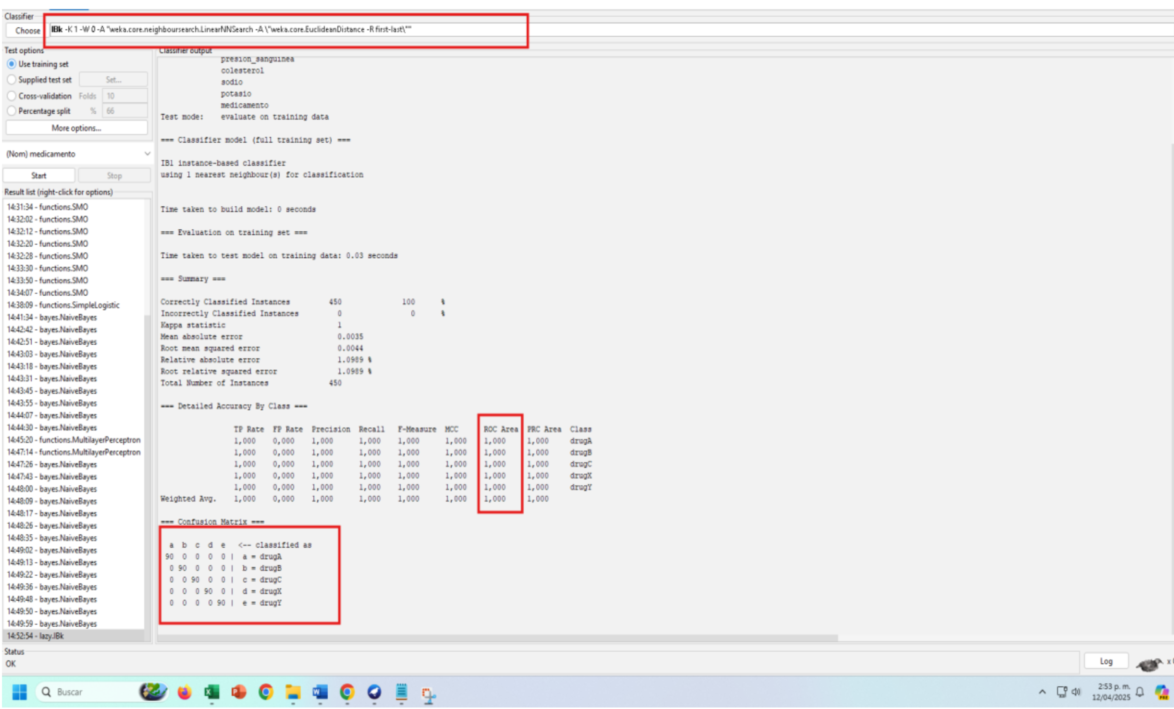
Método Bayesiano

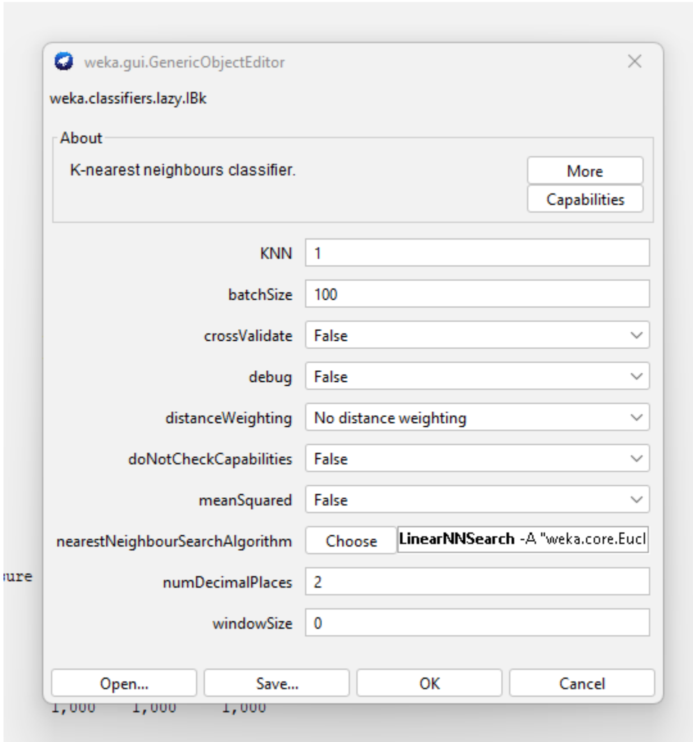


Configuración de caracterización



Método basado en ejemplos



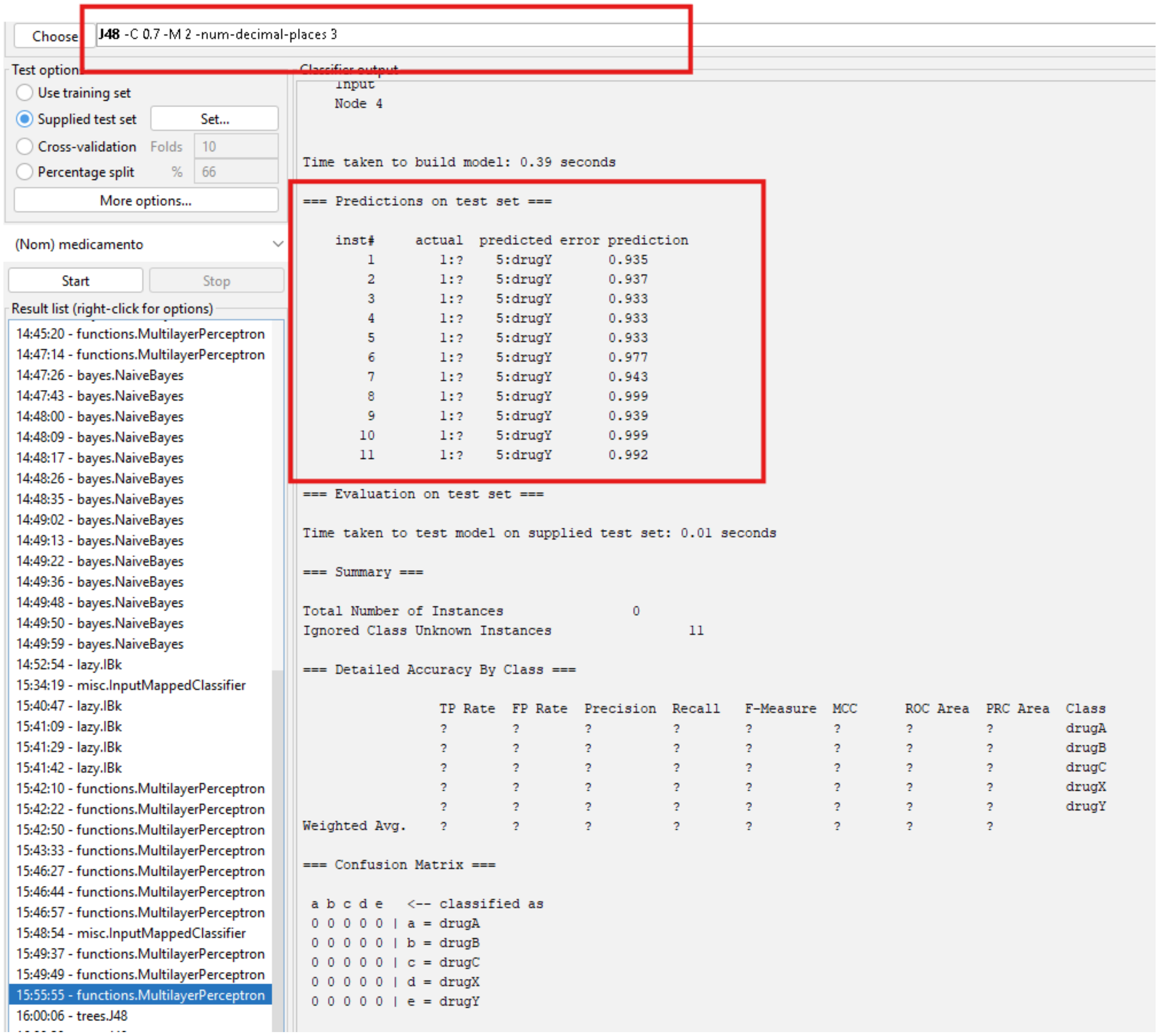


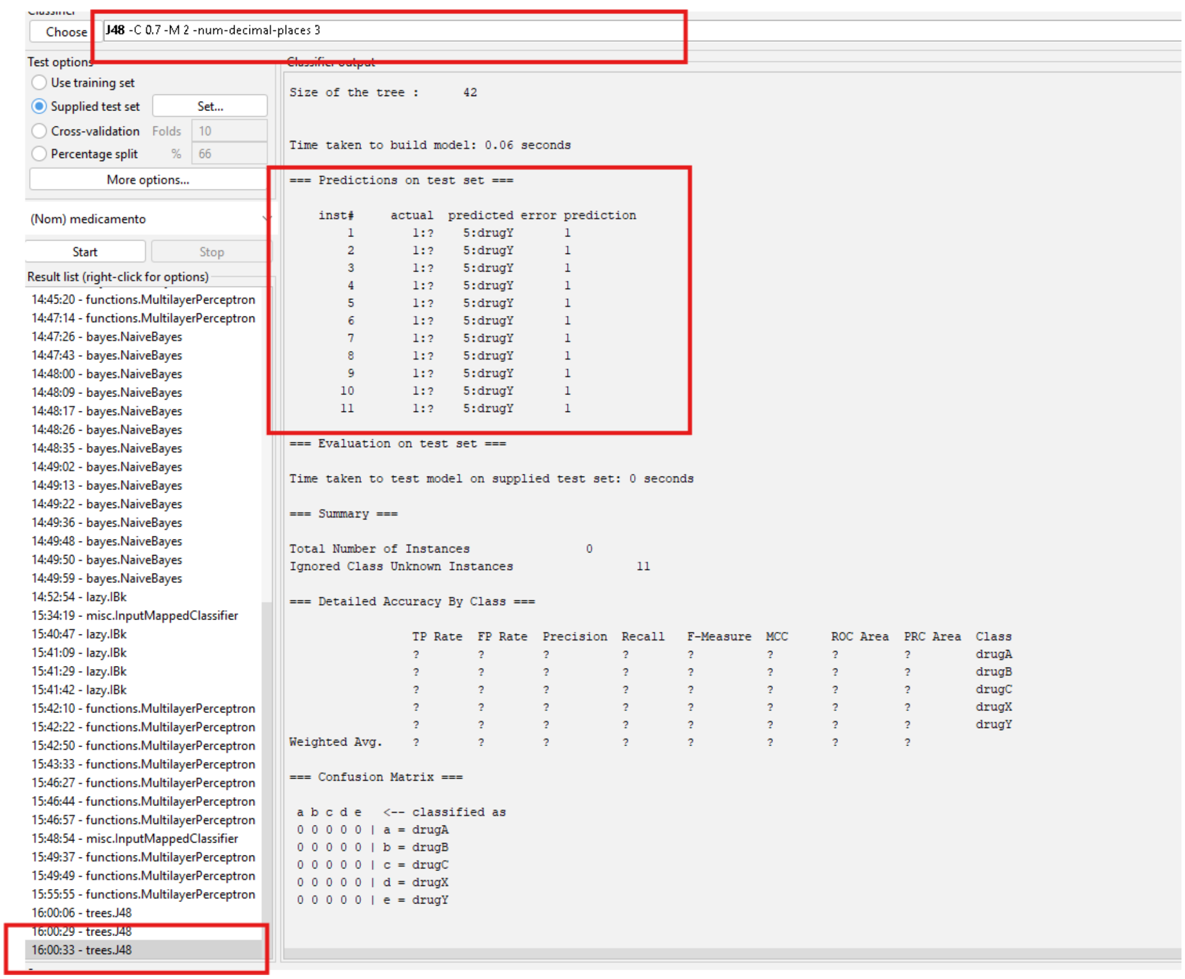
Series de Tiempo

NO aplica porque no tenemos fechas en la estructura

3. Luego de un exhaustivo análisis de los distintos modelos, tomamos la decisión de usar Redes neuronales porque es el método más robusto para solucionar al problema planteado y responde de manera correcta.

4. Luego de realizar la prueba con los nuevos datos de **pacientes_nuevos** encontramos que con el método de redes neuronales y árbol de métodos, el resultado de predicción es igual.


Choose
J48 -C 0.7 -M 2 -num-decimal-places 3
Test options
Use training set
Supplied test set
Set...
Cross-validation Folds 10
Percentage split % 66
More options...
(Nom) medicamento
Start
Stop
Result list (right-click for options)
14:45:20 - functions.MultilayerPerceptron
14:47:14 - functions.MultilayerPerceptron
14:47:26 - bayes.NaiveBayes
14:47:43 - bayes.NaiveBayes
14:48:00 - bayes.NaiveBayes
14:48:09 - bayes.NaiveBayes
14:48:17 - bayes.NaiveBayes
14:48:26 - bayes.NaiveBayes
14:48:35 - bayes.NaiveBayes
14:49:02 - bayes.NaiveBayes
14:49:13 - bayes.NaiveBayes
14:49:22 - bayes.NaiveBayes
14:49:36 - bayes.NaiveBayes
14:49:48 - bayes.NaiveBayes
14:49:50 - bayes.NaiveBayes
14:49:59 - bayes.NaiveBayes
14:52:54 - lazy.IBk
15:34:19 - misc.InputMappedClassifier
15:40:47 - lazy.IBk
15:41:09 - lazy.IBk
15:41:29 - lazy.IBk
15:41:42 - lazy.IBk
15:42:10 - functions.MultilayerPerceptron
15:42:22 - functions.MultilayerPerceptron
15:42:50 - functions.MultilayerPerceptron
15:43:33 - functions.MultilayerPerceptron
15:46:27 - functions.MultilayerPerceptron
15:46:44 - functions.MultilayerPerceptron
15:46:57 - functions.MultilayerPerceptron
15:48:54 - misc.InputMappedClassifier
15:49:37 - functions.MultilayerPerceptron
15:49:49 - functions.MultilayerPerceptron
15:55:55 - functions.MultilayerPerceptron
16:00:06 - trees.J48
Classifier output
Input
Node 4
Time taken to build model: 0.39 seconds
=== Predictions on test set ===
inst# actual predicted error prediction
1 1:? 5:drugY 0.935
2 1:? 5:drugY 0.937
3 1:? 5:drugY 0.933
4 1:? 5:drugY 0.933
5 1:? 5:drugY 0.933
6 1:? 5:drugY 0.977
7 1:? 5:drugY 0.943
8 1:? 5:drugY 0.999
9 1:? 5:drugY 0.939
10 1:? 5:drugY 0.999
11 1:? 5:drugY 0.992
=== Evaluation on test set ===
Time taken to test model on supplied test set: 0.01 seconds
=== Summary ===
Total Number of Instances 0
Ignored Class Unknown Instances 11
=== Detailed Accuracy By Class ===
TP Rate FP Rate Precision Recall F-Measure MCC ROC Area PRC Area Class
? ? ? ? ? ? ? ? drugA
? ? ? ? ? ? ? ? drugB
? ? ? ? ? ? ? ? drugC
? ? ? ? ? ? ? ? drugX
? ? ? ? ? ? ? ? drugY
Weighted Avg. ? ? ? ? ? ? ? ?
=== Confusion Matrix ===
a b c d e <-- classified as
0 0 0 0 0 | a = drugA
0 0 0 0 0 | b = drugB
0 0 0 0 0 | c = drugC
0 0 0 0 0 | d = drugX
0 0 0 0 0 | e = drugY


Choose
J48 -C 0.7 -M 2 -num-decimal-places 3
Test options
Use training set
Supplied test set
Set...
Cross-validation Folds 10
Percentage split % 66
More options...
(Nom) medicamento
Start
Stop
Result list (right-click for options)
14:45:20 - functions.MultilayerPerceptron
14:47:14 - functions.MultilayerPerceptron
14:47:26 - bayes.NaiveBayes
14:47:43 - bayes.NaiveBayes
14:48:00 - bayes.NaiveBayes
14:48:09 - bayes.NaiveBayes
14:48:17 - bayes.NaiveBayes
14:48:26 - bayes.NaiveBayes
14:48:35 - bayes.NaiveBayes
14:49:02 - bayes.NaiveBayes
14:49:13 - bayes.NaiveBayes
14:49:22 - bayes.NaiveBayes
14:49:36 - bayes.NaiveBayes
14:49:48 - bayes.NaiveBayes
14:49:50 - bayes.NaiveBayes
14:49:59 - bayes.NaiveBayes
14:52:54 - lazy.IBk
15:34:19 - misc.InputMappedClassifier
15:40:47 - lazy.IBk
15:41:09 - lazy.IBk
15:41:29 - lazy.IBk
15:41:42 - lazy.IBk
15:42:10 - functions.MultilayerPerceptron
15:42:22 - functions.MultilayerPerceptron
15:42:50 - functions.MultilayerPerceptron
15:43:33 - functions.MultilayerPerceptron
15:46:27 - functions.MultilayerPerceptron
15:46:44 - functions.MultilayerPerceptron
15:46:57 - functions.MultilayerPerceptron
15:48:54 - misc.InputMappedClassifier
15:49:37 - functions.MultilayerPerceptron
15:49:49 - functions.MultilayerPerceptron
15:55:55 - functions.MultilayerPerceptron
16:00:06 - trees.J48
16:00:29 - trees.J48
16:00:33 - trees.J48
Size of the tree : 42
Time taken to build model: 0.06 seconds
=== Predictions on test set ===
inst# actual predicted error prediction
1 1:? 5:drugY 1
2 1:? 5:drugY 1
3 1:? 5:drugY 1
4 1:? 5:drugY 1
5 1:? 5:drugY 1
6 1:? 5:drugY 1
7 1:? 5:drugY 1
8 1:? 5:drugY 1
9 1:? 5:drugY 1
10 1:? 5:drugY 1
11 1:? 5:drugY 1
=== Evaluation on test set ===
Time taken to test model on supplied test set: 0 seconds
=== Summary ===
Total Number of Instances 0
Ignored Class Unknown Instances 11
=== Detailed Accuracy By Class ===
TP Rate FP Rate Precision Recall F-Measure MCC ROC Area PRC Area Class
? ? ? ? ? ? ? ? drugA
? ? ? ? ? ? ? ? drugB
? ? ? ? ? ? ? ? drugC
? ? ? ? ? ? ? ? drugX
? ? ? ? ? ? ? ? drugY
Weighted Avg. ? ? ? ? ? ? ? ?
=== Confusion Matrix ===
a b c d e <-- classified as
0 0 0 0 0 | a = drugA
0 0 0 0 0 | b = drugB
0 0 0 0 0 | c = drugC
0 0 0 0 0 | d = drugX
0 0 0 0 0 | e = drugY